



What makes a K-pop Idol Instagram Famous?

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Abstract

As K-pop expands globally, questions arise about what drives certain idols to gain massive followings while others remain unknown. What traits matter most, and does beauty play that big of a role? Further, can ChatGPT analyze an idol's perceived beauty in a meaningful way to correlate with their Instagram following? This study analyzes a dataset of 366 idols to evaluate how demographic, physical, and career-related factors predict Instagram popularity. Using XGBoost, OLS, Ridge, Random Forest, and MLP models, we measure how each feature contributes to follower count. Although physical characteristics have subtle influence, company affiliation and Spotify followers consistently dominate across methods. Factors that are important in determining idol success are analyzed and it is concluded that the "halo effect" a K-pop artist may have is minimal, contrary to popular belief.

Introduction

- K-pop has grown into a global phenomenon with an estimated 150–200 million [5] international fans, many of whom are adolescents prone to **identity seeking, and parasocial attachment** [2].
- **Korean Beauty Standards**, play a major role in how idols are perceived. K-pop idols' average BMIs often fall in the underweight range, reflecting strict standards that fuel the halo effect where attractiveness is linked to perceived talent and likeability [1].
- **Social Identity Theory** helps explain fandom behavior: fans form strong in-groups based on nationality, company, or fandom affiliation, contributing to loyalty and rapid online engagement [3].
- Beyond an idol's individual characteristics, there are various **exterior factors** contributing to their success, such as the group they are in or what company they are from [4].
- On platforms like Instagram, where top idols regularly exceed 10 million followers, psychological and structural forces merge, making follower count an indicator of how identity, beauty norms, and industry power shape idol popularity.

Methods

- Downloaded, merged, and cleaned datasets from **Kaggle** to gain large net of information about 366 individual idols
- Scraped through **Spotify API** to get number of followers of each idol's kpop group
- Made two different "rankings" of idols in dataset from most to least attractive according to Korean Beauty Standards
 - Folders of ~30 images of each idol were created based off of Bing search of their Korean stage name and Korean group name
 - First, "trained" rankings were based on a **ChatGPT** trained on images of "Top," "Mid," and "Low" beauty rankings per gender based on netizen's preferences.
 - Second, "untrained" rankings were based on ChatGPT going through each folder of idols and rating them based on its own, pre-existing knowledge of beauty standards.
- Used **SHAP analysis** to determine which features to discard and which to keep in training models
- Altered variables and parameters in models until reaching a 77% accuracy in **OLS and Ridge Regressors**

Finding 1: Group Fame is the Top Predictor of Spotify Followers

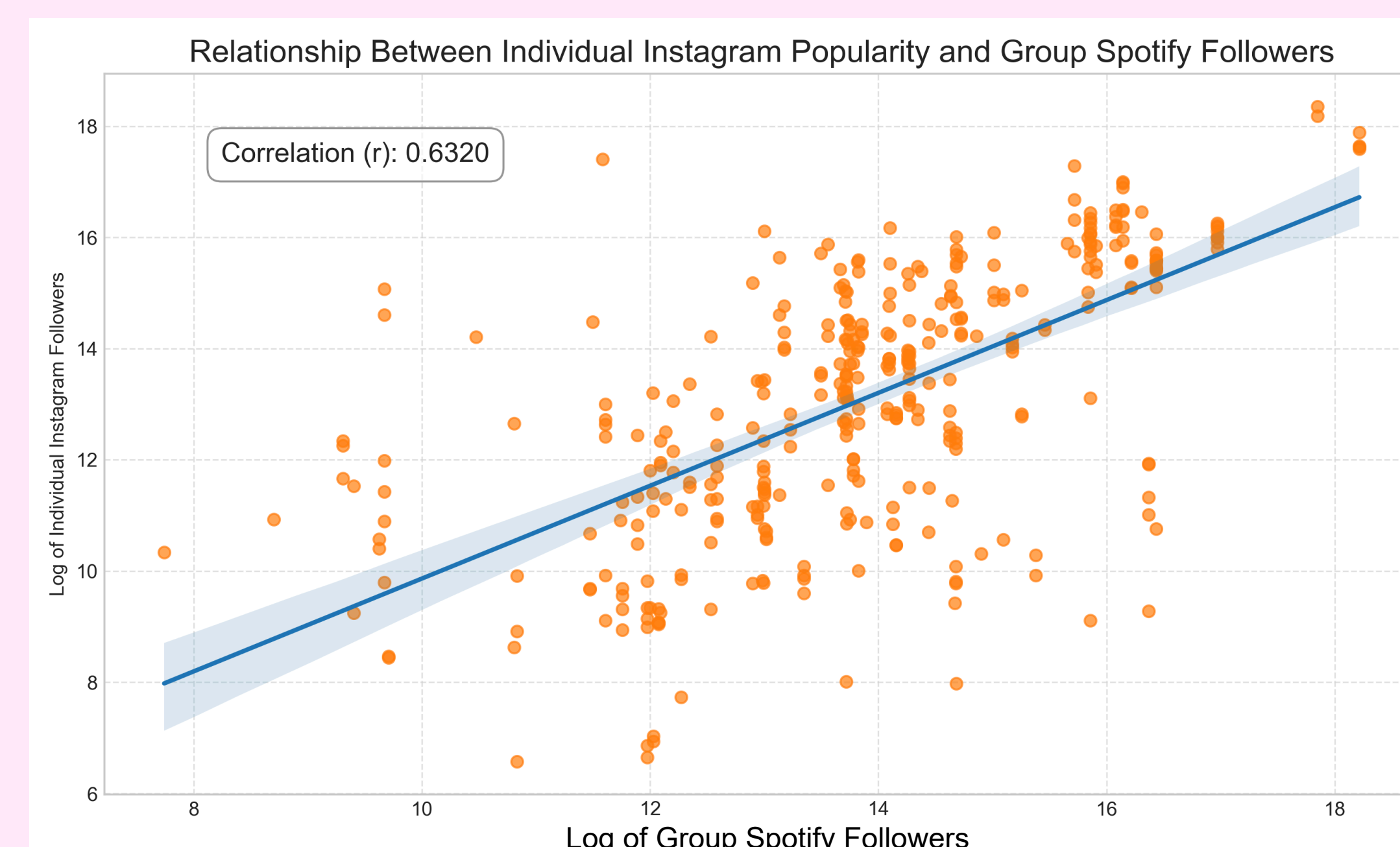


Figure: Spotify followers of an idol's group compared to followers of an individual. Adjusted to be logged to factor for outliers.

Results from a SHAP analysis indicate that the followers of an idol's group on Spotify is the strongest predictor of how many Instagram followers they have. The model demonstrates how much of an indicator being part of a well known, highly listened to group determines how many Instagram followers an individual idol has. The result shows a relevant, positive correlation between the two factors, indicating that group success is necessary for that of an individual idol.

Finding 3: ChatGPT trained on hand-selected Images outperforms the base model

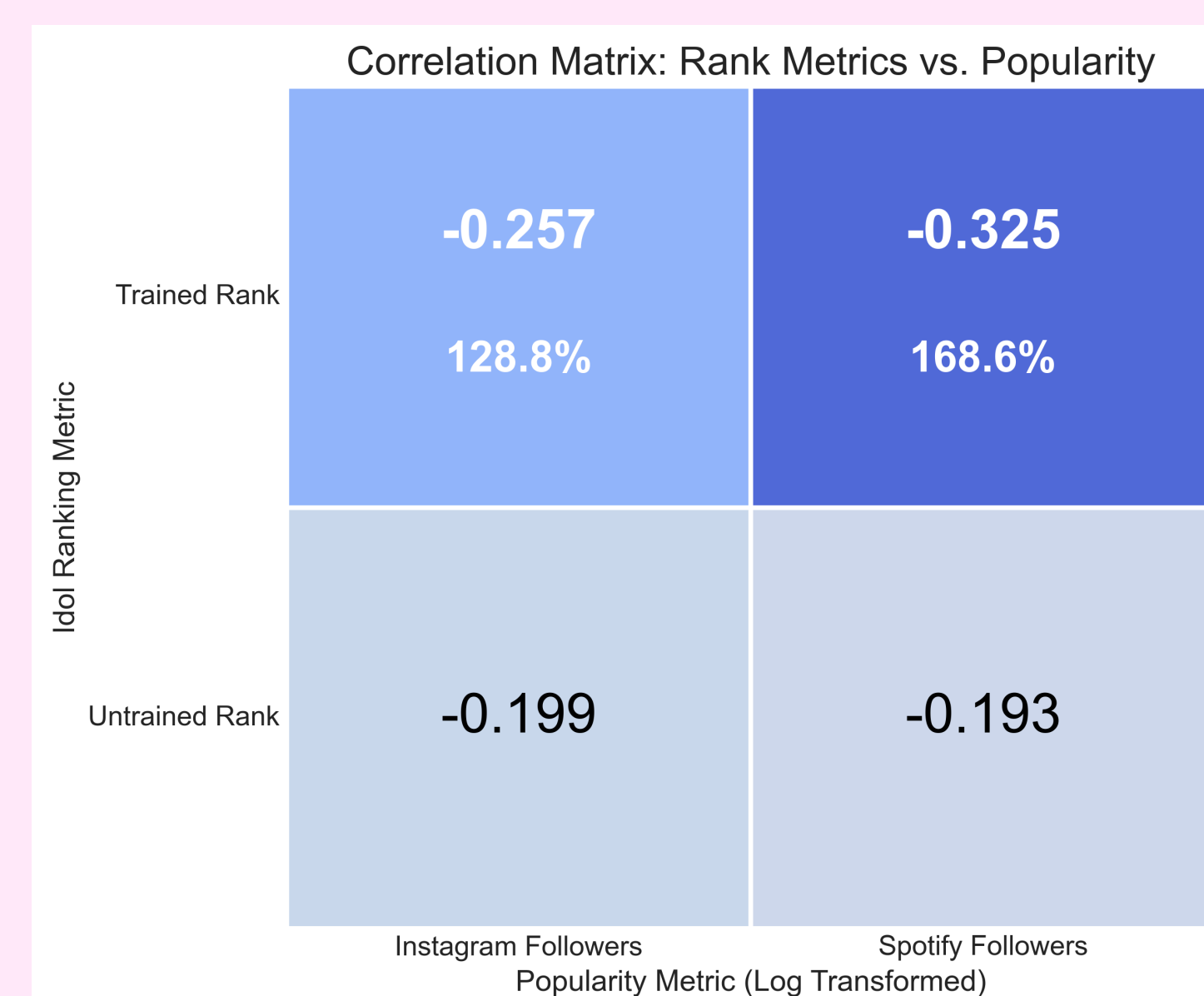
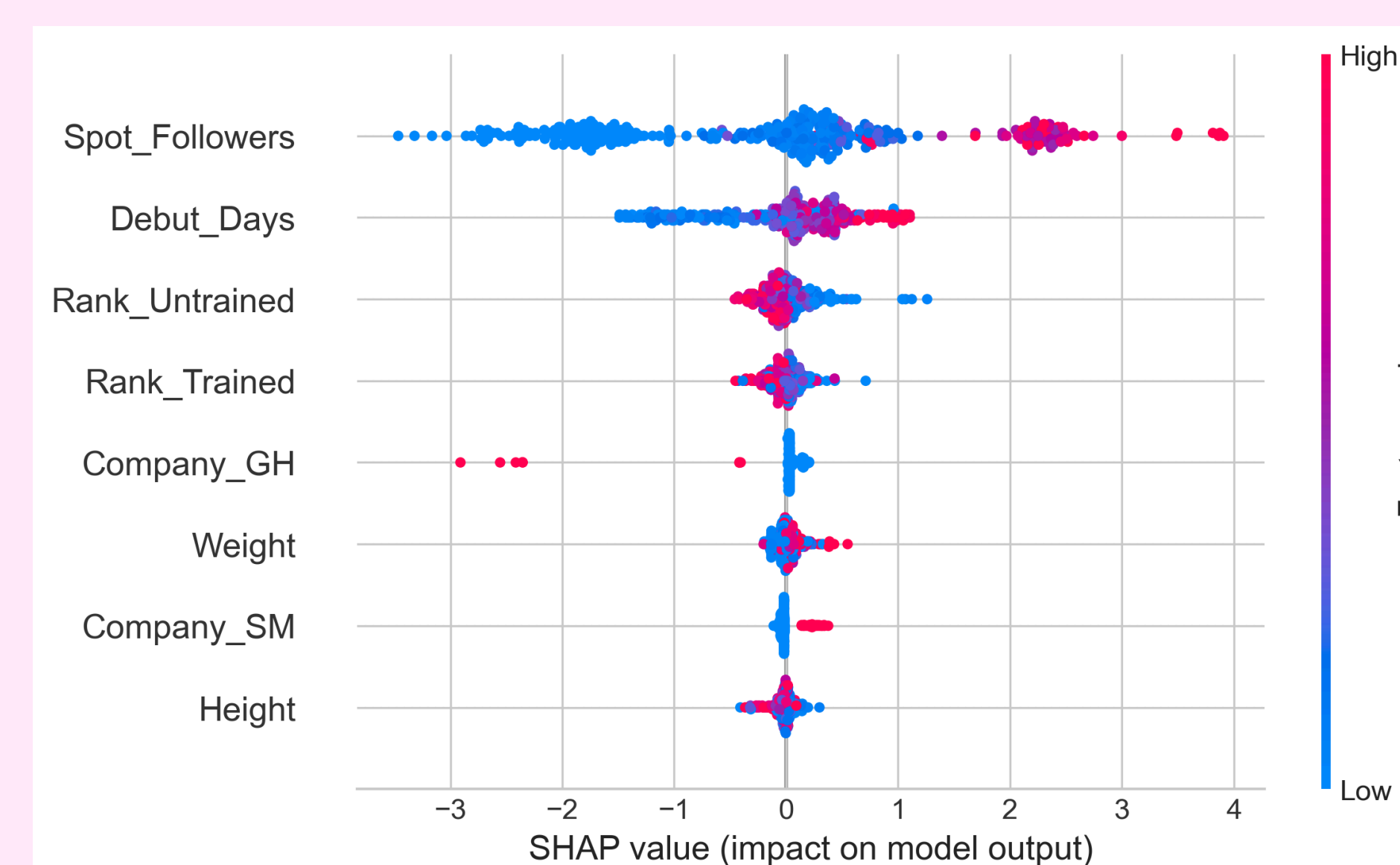


Figure: Each idol got two different rankings from ChatGPT: one version that was trained on images demonstrating perceived beauty of kpop idols, and one based on its own, preexisting knowledge.

Top 7 Predictors determined by XGRegressor SHAP Analysis



Finding 2: Agency Power Predetermines an Idol's Fate

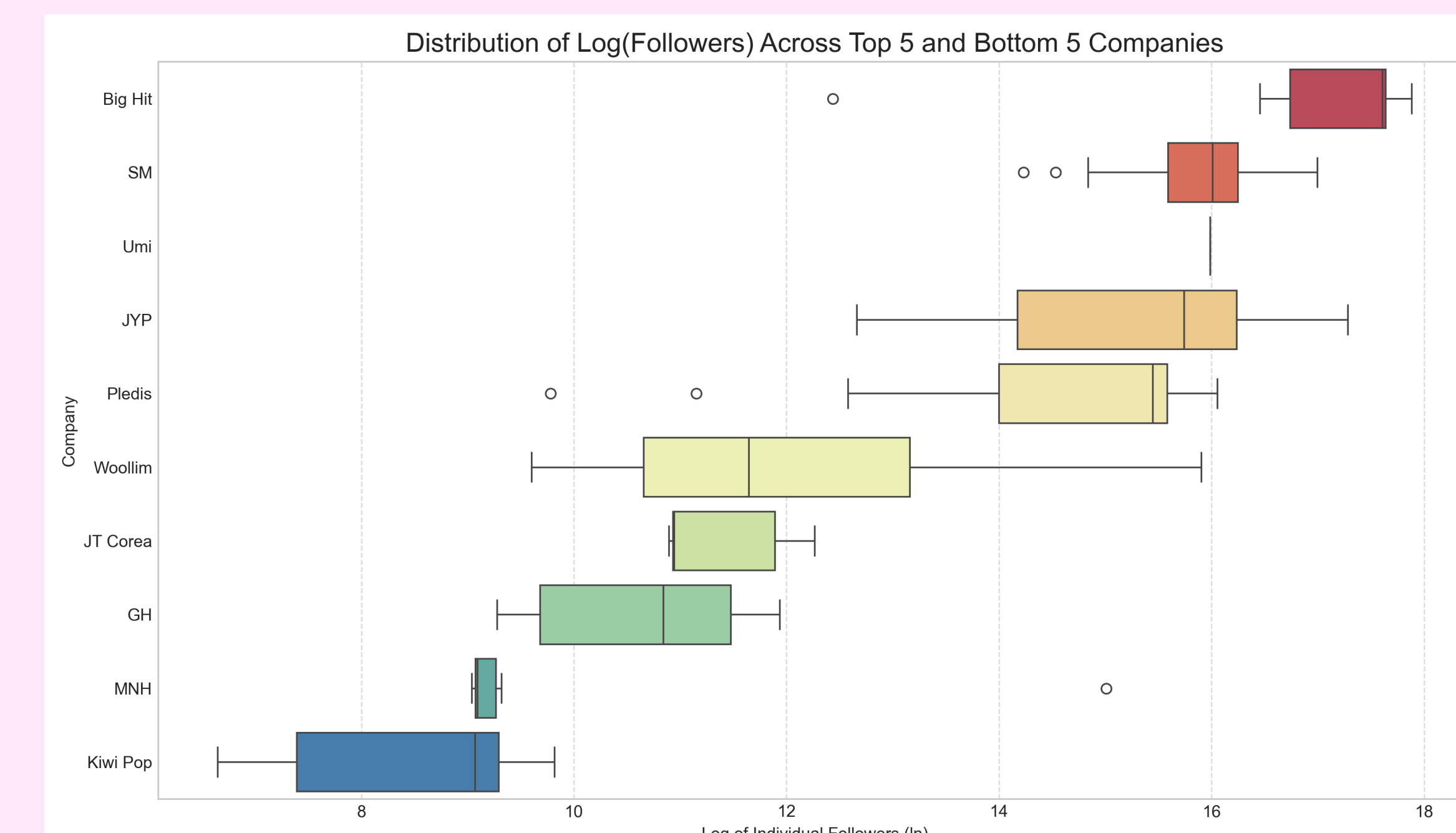


Figure: Top 5 and bottom 5 companies compared against Instagram followers (logged to account for outliers) in accordance to amount of idols they manage as a histogram



Top companies have medians that are significantly higher than the absolute maximum reached by idols in the bottom tier. An idol debuting at a Top 5 company, benefits from a vastly higher starting 'floor' of popularity. Even the least followed idol at a top company often has more followers than the average or best-performing idol at a bottom company. Bottom-tier companies show a constrained distribution, indicating a limit to the popularity an individual idol can achieve when not backed by the resources of a major agency.

Conclusion

- Appearance plays a nearly insignificant role in predicting an idol's popularity compared to structural factors such as group belonging or company affiliation.
- While BMI can affect attractiveness rankings, as an individual factor, it, along with weight, has virtually no effect on an idol's popularity in Instagram following and Spotify followers.
- ChatGPT does not have a complete understanding on what makes an idol more visually alluring nor how to compare one to Korean Beauty Standards.
- Idols mainly rely on outer sources as predictors of fame and popularity rather than individual characteristics.



References

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